



2026-27

CONCEPTS | CLINICALS | MCQS

PHARMACOLOGY

- Drugs Simplified.
- Questions Demystified.
- Tables, classifications & side-effects made easy.
- PYQs + tricky options decoded.



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PHARMACOLOGY

Pharmacokinetics

Effect of Body on Drug aka ADME Study

- Absorption
- Distribution
- Metabolism
- Excretion

Absorption → It is the movement of drugs from the site of administration to blood.

→ Lipid Solubility is the most important factor in absorption.

Acidic Drug [Aspirin] → mainly absorbed from stomach

Basic Drug [Morphine] → mainly absorbed from intestine

When The Medium Is Same, Then The Drug Will Cross.

Bioavailability

It is the fraction of given dose that reaches the systemic circulation in unchanged form

Bioavailability : Determines the Dose

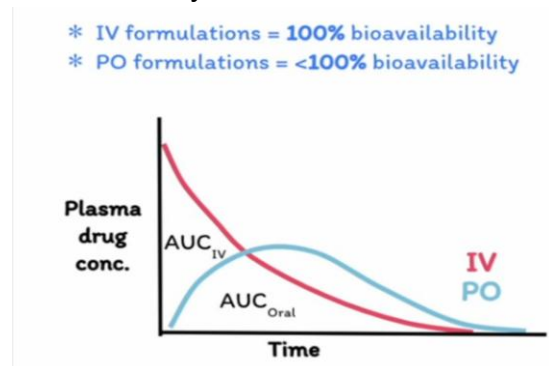
High bioavailability → Low Dose

Low bioavailability → High dose

Factors affecting bioavailability :

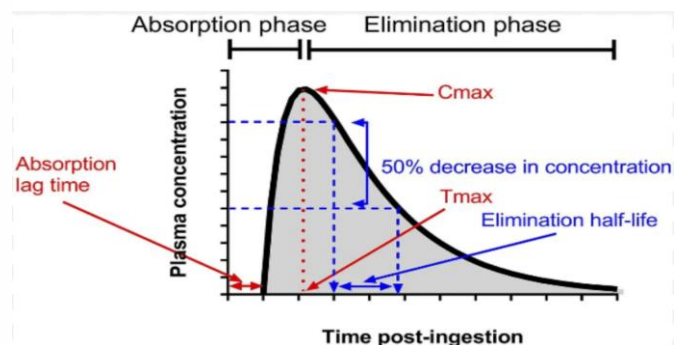
1. Absorption
2. Route of administration
3. First Pass metabolism

Bioavailability



$$\text{Bioavailability} = \frac{\text{AUC}_{\text{Oral}}}{\text{AUC}_{\text{IV}}} \times 100$$

Plasma concentration vs time graph



•Frequently asked points ✓

- 1)C_{max} : Maximum concentration obtained by a **particular dose**
- 2)T_{max}: Time in which plasma concentration becomes maximum -Tells **Rate of absorption**
- 3)AUC: Total area covered by graph- Tells **Extent of absorption**

Volume of Distribution (V_d)

V_d- amount of drug in body/concentration of drug in plasma

V_d ∝ Amount of Drug in Tissues

More V_d = More distribution

Loading dose-

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Initial high dose given to start the action

$$LD = V_d \times \text{Target Plasma Concentration}$$

Maintenance dose-

Repeated doses given to maintain the plasma concentration.

$$MD = CL \times \text{Target PC}$$

Drug which can be removed by dialysis should be →

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- 1.Low volume distribution
- 2.Low plasma protein binding

Drugs can be removed by dialysis

Methanol

Lithium

Aspirin

Valproic acid

Metformin

Can not removed by dialysis (AVOID)

Amphetamine

verapamil

opioids

imipramine

digoxin/diazepam

Metabolism

Aim of Metabolism → To Make a Drug Water Soluble (Polar)

Phase I reactions→ CHORD

Cyclization

Hydrolysis

Oxidation (most common)

Reduction

Deamination

Phase II reactions

Glucuronide conjugation

Glutathione conjugation

Acetylation

Methylation

Purpose of Phase I → Expose the Functional Group on the drug

Purpose of Phase II → Makes the drug water soluble

Enzyme inducers	Enzyme inhibitors
↓ the metabolism of drugs metabolised by microsomal enzymes	↑ the metabolism of drugs metabolized by microsomal enzymes
This increases plasma concentration and can lead to toxicity.	This reduces plasma concentration and thus action of these drugs
Mnemonic- GPRS cell phone G- griseofulvin P- phenytoin R- rifampicin S- smoking Cell- carbamazepine Phone- phenobarbitone	Tricks- Vit - valproate K- ketoconazole Can't- cimetidine Cause - ciprofloxacin Enzyme - erythromycin Inhibition- isoniazid

Drugs metabolized by CYP 3A4 (CT SCAN)

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C- Cyclosporine, CCB, carbamazepine

T- Tacrolimus

S- Statins

C- Cisapride Astemizole Terfenadine (CAT drugs → QT prolongation in ECG)

A- Amiodarone

N- Navir (HIV protease inhibitors)

Drugs metabolised by CYP 2C19

Clopidogrel

Ticlopidine

Drugs metabolized by 2C9

Clotting antagonist (Warfarin)

Phenytoin

Clearance - Rate of elimination/ plasma concentration

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Half Life [t_{1/2}]

It is the time in which plasma concentration of a drug becomes half.

Half-life is constant in first order kinetics.

Importance of t_{1/2}:

- Dose cannot be calculated
- Dosing Interval/ Frequency can be calculated

$$t_{1/2} = \frac{0.693 \times V_d}{CL}$$

Diagram labels for the equation above:

- Logarithm of 2 (points to 0.693)
- Volume of distribution (points to V_d)
- Clearance (points to CL)
- Half-life (points to t_{1/2})

Order of kinetics

First order kinetics	Zero order kinetics
Fraction is constant	Amount is constant
Rate of elimination $\propto$ Plasma concentration	Rate of elimination = Constant
CL = Constant t _{1/2} = Constant	CL $\propto$ 1/PC t _{1/2} $\propto$ PC

Drugs Following Zero Order Kinetics

ZERO

W → Warfarin

A → Alcohol/Aspirin

T → Theophylline

T → Tolbutamide

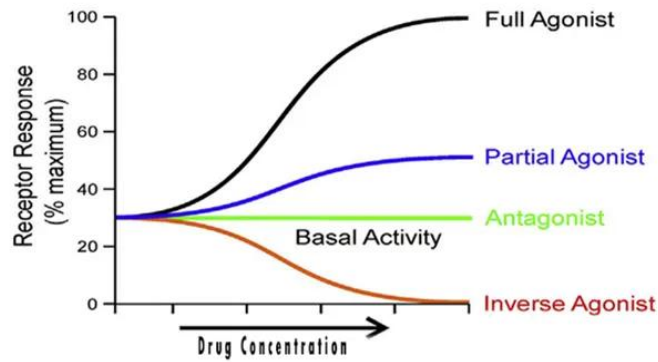
Power → Phenytoin

Pharmacodynamics:

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Classification of Drugs Based on Intrinsic Activity

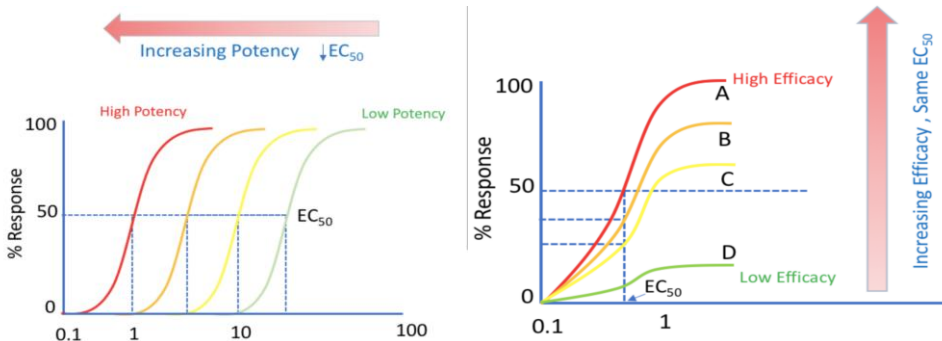
Agonist	Maximum intrinsic activity
Partial agonist	Submaximum intrinsic activity
Inverse agonist	Opposite action to agonist
Antagonist	No action of itself but interfere with action of other drugs

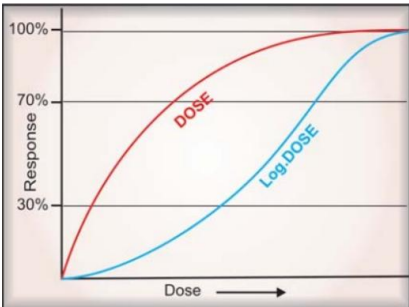


Potency → To produce a similar response if a drug is required in lesser dose, it is more potent → On log DRC, left-sided curve is more potent drug whereas right-sided curve is less potent

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Efficacy → Relates to maximum effect of a drug regardless of dose
→ Higher the curve, more is efficacy



Dose response curve	Log dose response curve
Graph is hyperbola	Sigmoid curve
 The graph shows two curves on a plot of Response (30%, 70%, 100%) vs Dose. The red curve, labeled 'DOSE', is a hyperbola. The blue curve, labeled 'Log-DOSE', is a sigmoid curve. The x-axis is labeled 'Dose' with an arrow pointing right.	3 imp parameters obtain- Efficacy Potency slope

Quantal DRC

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THERAPEUTIC INDEX - LD50/ED50 (trick- TILE)

ED50 - dose at which 50 percent of people get desired response

LD50- lethal dose at which half of the people died

Therapeutic index is margin of the safety of a drug

- If ↑ Therapeutic index-drug is safe
- If ↓ Therapeutic index-drug is unsafe.

CLINICAL TRIALS

PHASE I	PHASE II	PHASE III	PHASE IV
To check maximum tolerable dose Evaluate efficacy	Check safety of drug	For confirmation of efficacy	To check rare and adverse side effects
Done on 20-50 people	Done on 200-500 people	Done on 2000-5000	Unpredictable
Done on healthy	Done on patient	Done on patient	Done on patient
Phase I can also be done in Patients for Toxic drugs	Indicator of Efficacy [1st time efficacy is done]		
Aim of Phase I trials: MTD > Dosing > PK > Safety	Maximum risk of failure is in phase II		

Pharmacogenetics

A. Glucose-6 Phosphate Dehydrogenase Deficiency

→ G-6 PD protects RBC from free radical injury

→ Many drugs produce free radicals and can cause hemolysis in patients with genetic deficiency of G-6-PD

Drugs causes hemolysis in G6PD deficiency-

Primaquine, Sulfonamides, Nitrofurantoin, fluoroquinolones, nalidixic acid

B. Acetylation

→ Drugs metabolized by acetylation produce different effects in different patients.

S→ Sulfonamide [including Dapsone]
H→ Hydralazine
I→ INH
P→ Procainamide
[SHIP Drugs can cause SLE also]

Teratogenic Effect of Drugs

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Warfarin	Nasal hypoplasia Calcific stippling of epiphyses Hearing loss, purple toe syndrome
Valproate	Neural tube defects
Thalidomide	Phocomelia
Tetracyclines	Teeth and bone problems
Fluoroquinolones	Cartilage and tendon damage
Phenytoin	Fetal hydantoin syndrome
Lithium	Ebstein's anomaly
Carbimazole	Aplasia cutis & Choanal atresia

Enteric Coating of Drugs:–

This is a coating that dissolves in alkaline medium only.

– It can be used for

- Protecting the acid labile drugs from gastric acid, e.g., proton pump inhibitors.
- It can protect gastric mucosa from irritant drugs.
- It increases the absorption of drugs that are preferentially absorbed distal to the stomach.

Schedule H:– Drugs that require prescription from a registered medical practitioner are called Schedule H drugs.

AUTONOMIC NERVOUS SYSTEM

1.Cholinergic Drugs

1.Cholinergic Receptors

Nicotinic Receptors

a.NN → Ganglia

b.NM → NMJ (require optimal stimulation)

Muscarinic Receptors

M1 → Stomach: ↑HCl

M2 → Heart: ↓ HR, ↓ conduction

M3 → Bronchus ,GIT ,Bladder ,Glands ,Pupil

Directly acting cholinergic drugs:

1.Pilocarpine- act on pupil → miosis

Given in → Angle closure glaucoma

2.Bethanechol- act on urinary bladder → ↑ urinary outflow

Given in → Atonic Bladder

3.Methacholine - act on heart → ↓ HR

Indirectly acting cholinergic drugs:- These drugs act by inhibiting AChE

1. Physostigmine- use in ACG (Acute angle glaucoma)

DOC for atropine poisoning.

2. Neostigmine and Pyridostigmine use for Myasthenia Gravis

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Tensilon test (**Edrophonium test**)- use for Myasthenia Gravis

Imp note-

Directly acting cholinergic drug used to treat glaucoma → pilocarpine

Drug used for dryness in sjogren syndrome → cevimeline

Myasthenia Gravis treatment -

→ Neostigmine or Pyridostigmine are used.

→ Pyridostigmine longer acting than neostigmine.

Neostigmine

Nicotinic uses	Muscarinic uses
Myasthenia gravis Cora bite	Post operative paralytic ileus Post operative urinary retention

Muscle relaxant reversal	
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ORGANOPHOSPHATE POISONING-

D- Diarrhea
U- Urination
M- Miosis
B- Bradycardia
B- Bronchospasm
E- Emesis
L- Lacrimation
S- Salivation

Frequency asked pyq → DOC- ATROPINE

Most reliable sign of atropinization- ↓ secretions

Atropine can reverse all symptoms of opioid poisoning except muscle weakness

ANTICHOLINERGIC DRUGS-

Drugs given in acidity(M1)- pirenzepine, telenzepine

ATROPINE-

1. Heart (M2) for bradycardia
2. Glands(M3) ↓ secretions

Atropine poisoning → Symptoms →

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D- Dry mouth
H- Hyperthermia
A- Agitation
T- Atropine
U- Urinary
R- Retention
A- Accommodation

DOC - Physostigmine

M3 blockers → causes bronchodilation
Ipratropium, tiotropium

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→ act on urinary bladder

Given in urinary incontinence, overactive bladder

Solefenacin, oxybutynin, flavoxate, tolterodine, darifenacin

SCOPOLAMINE- for motion sickness prophylaxis

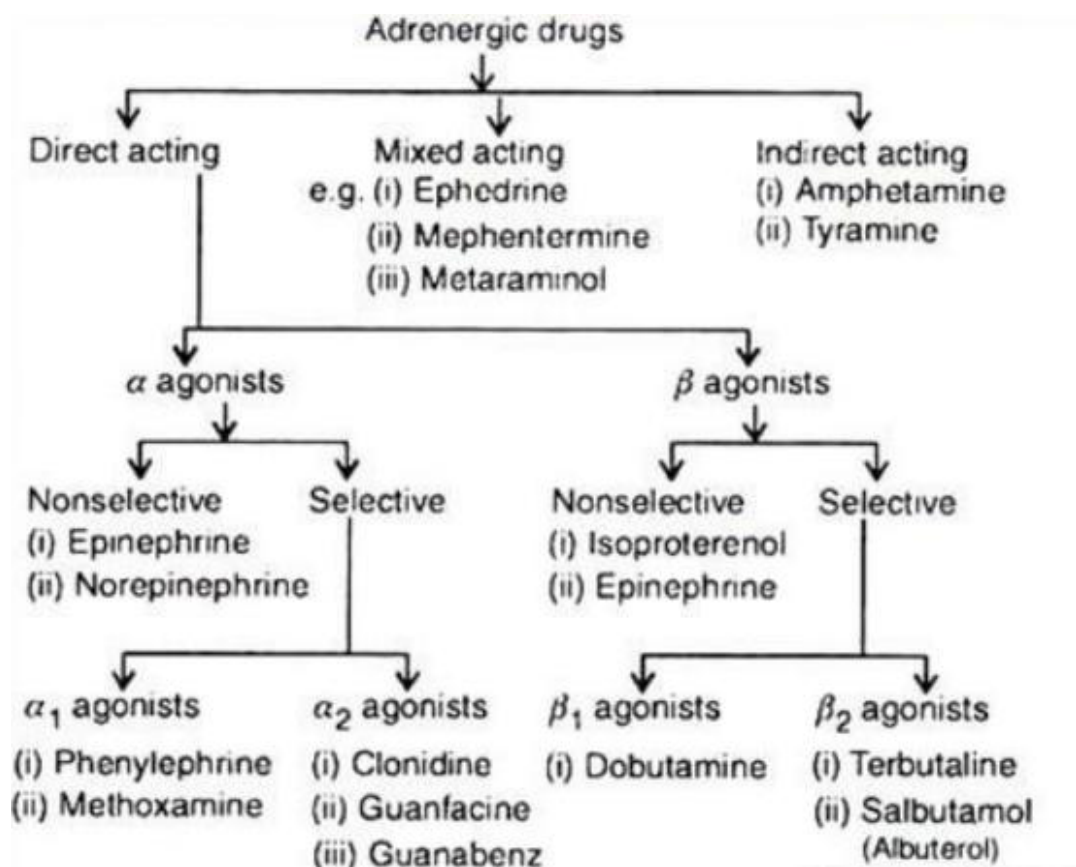
Note- Acetazolamide- for mountain sickness

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GLYCOPYRROLATE- ↓ secretions

ADRENERGIC DRUGS-

α_1	α_2	β_1	β_2
Bloodvessel - Vasoconstriction	Presynaptic - act as brake	Heart	GIT
Eyes- mydriasis		JG cells of kidney	Lungs
Prostatic urethra - ↓ urine outflow			Urinary bladder Uterus Blood vessel - vasodilation



COCAINE – It is a local anaesthetic

- Cocaine cause vasoconstriction and hypertension
- Shows tactile hallucinations (cocaine bugs)
- Cause palatine perforation

Dopamine -Acts as dose dependent

D1	<2 mcg/kg/min	Blood vessels	Vasodilation
β1	2-10 mcg/kg/min	Heart	Tachycardia
α1	>10 mcg/kg/min	Blood vessels	vasoconstriction

DOC in shock + oliguria

Fenoldopam → use in HTN emergency + oliguria

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Drugs	Systolic BP β1	Diastolic BP α1 β2	Heart -Direct acting β1	Heart -Indirect acting Opposite to DBP	Net effect
Adrenaline	↑	-	↑	-	↑
Nor adrenaline	↑	↑↑	↑	↓↓	↓
Isoprenaline	↑	↓↓	↑	↑↑	↑↑↑

ADRENALINE-

1. ANAPHYLACTIC SHOCK- DOC is Adrenaline – Route of choice is I.M. or S.C.

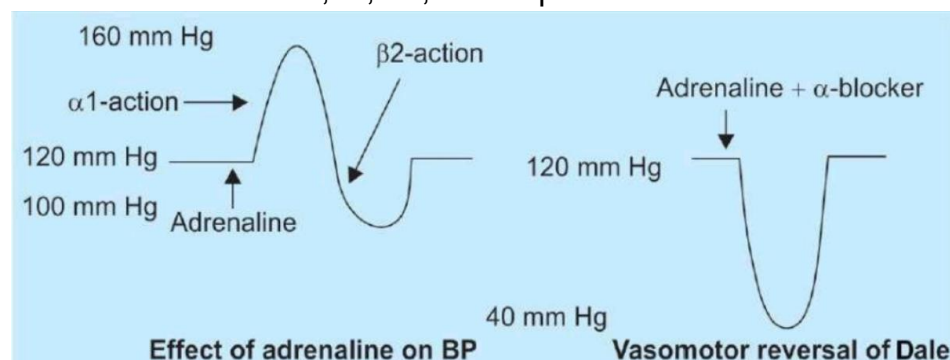
Concentration is 1:1000 (0.5 ml)

2. CARDIAC ARREST – Adrenaline is useful – Concentration is **1:10,000** –Preferred routes are IV

3. LOCAL ANESTHETICS – Adrenaline is added with LA to ↓ systemic toxicity – Concentration is **1:1,00,000 – 1:2,00,000**

VASOMOTOR REVERSAL OF DALE

Adrenaline acts on α1,α2, β1,β2 receptors.



α1- Vasoconstriction β2- Vasodilation

Biphasic response if ADR is given

When adrenaline given with alpha blocker- fall of BP occurs

So α blockers alone avoided in pheochromocytoma which is producing adrenaline

ANTI- ADRENERGIC DRUGS

1. ALPHA BLOCKERS

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NON- SELECTIVE	SELECTIVE
PHENOXYBENZAMINE- use for pheochromocytoma to prevent catecholamine crisis PHENTOLAMINE - use for cheese reaction and for clonidine withdrawal.	PRAZOSIN, TERAZOSIN- use in BPH *If BPH patient is complaining of orthostatic hypotension after taking prazosin replace with tamsulosin PRAZOSIN doc - for scorpion sting

2. BETA BLOCKERS

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β_1 blockers are cardio selective and can be used in Asthma,DM.

Cardioselective Beta Blockers

Metoprolol
Atenolol
Nebivolol
Bisoprolol
Acebutolol
Betaxolol
Esmolol



Longest acting β blockers- Nadolol

Short acting- Esmolol metabolized by Pseudocholinesterase

β_2 blockers given in -

- Performance anxiety
- Thyrotoxicosis
- Glaucoma
- Akathisia
- Migraine prophylaxis
- Essential tremors

β_1 blocker is CI in acute CHF

3. PDE-5 INHIBITORS-

Sildenafil, Vardenafil, Tadalafil

Uses- Erectile dysfunction, Pulmonary hypertension

ALPHA AGONISTS

β 2	Drug	Uses
	SALBUTAMOL TERBUTALINE SALMETEROL FORMETEROL	Bronchodilator Used for Asthma & COPD Side effects - hyperglycemia,hypokalemia
	RITODRINE ISOXSUPRINE.	For preterm labour

↑ trabecular outflow	↑ uveoscleral outflow
Miotics - pilocarpine Netarsudil	PGF2α - latanprost, bimatoprost

Adverse Effects

MIOTICS	→ Cataract → Stenosis of NLD
PGF2a Analogues	→ Pigmentation of Iris (Heterochromia Iridis) → Growth of eyelashes (Hypertrichosis) → Fluid in macula (Macular edema)
Apraclonidine	→ Lid retraction
Brimonidine	→ Causes Brain suppression in Infants (Leads to Apnea) – C/I in children < 2yrs
Epinephrine	→ It is metabolized to form Adrenochrome - Causes black pigmentation of Conjunctiva
Netarsudil (Rho-kinase inhibitor)	→ Conjunctival hyperemia– Cornea verticella

Migraine

NEETPG 25,22,FMGE 25,24

Treatment of acute attack:

- Drug of choice for mild to moderate headache → **NSAIDs (Paracetamol, diclofenac)**
- Drug of choice for Acute severe attack → **Triptans**
- Other drug for Acute severe migraine → **Ergotamine**
- Because of side effects (increased vomiting and gangrene); Triptans are preferred over ergotamine.
- Both triptans and ergotamine should never be given together.
- Both can cause vasoconstriction resulting in coronary artery spasm, so avoided in patients with coronary artery disease.
- For prophylaxis → **Propranolol**

New Drugs for Migraine:

1. Lasmiditan (DITANS)

- Stimulates 5HT_{1F} receptor and decreases CGRP release.
- This prevents vasodilation and neuronal inflammation.

2. Monoclonal Antibodies Against CGRP

- Erenumab
- Fremanezumab

3. CGRP Antagonist: (GEPANTS)

- Olcegepant
- Rimegepant
- Ubrogepant

HISTAMINE

H₁ -CNS, Bronchus

H₂ -Stomach

H1 blockers -

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1st generation	2nd generation
Cross BBB causes sedation.	Do not cross BBB, no sedation
Have anticholinergic S/E	No anticholinergic action
Ex: Promethazine, Diphenhydramine, Pheniramine.	Terfenadine, Astemizole, Fexofenadine Cetirizine, levocetirizine

5-HT Serotonin Receptors

NEETPG 25. NEETPG 22

Frequently asked points✓

5HT1 Agonist - Buspirone uses- **Anxiety**

5HT1B/D Agonist - Ergotamine, Sumatriptan uses- **Acute severe migraine**

5HT3 (Blocker)- Ondansetron, Granisetron, Tropisetron, Dolasetron, Palonosetron- **use for chemo induced nausea/vomiting**

5HT4 (Agonist)-Cisapride, Mosapride, Prucalopride- GERD act as prokinetic

Metoclopramide →

prokinetic agent act as dopamine D2 receptor antagonist and 5-HT4 receptor agonist
Can cause extrapyramidal side effects

ALCOHOL WITHDRAWAL

Frequently asked points✓

Delirium tremens → IV benzodiazepines (diazepam)

If liver disease → Lorazepam or Oxazepam

Korsakoff psychosis → Antipsychotics given and for long term treatment give thiamine

Maintenance to prevent another episode of withdrawal → DOC - Chlordiazepoxide

Anticraving → Naltrexone (opioid antagonist) + Acamprosate (NMDA antagonist)

PGs and NSAIDs

Organ

Action

1) General

– Fever, Pain, Inflammation

2) Platelets

– TXA2 → Aggregation

– PGI2 → Inhibition of aggregation

3) Heart

– Ductus Arteriosus is kept open by PGE1

Patent ductus Arteriosus is treated by NSAIDs like

• Aspirin • Indomethacin • Ibuprofen (DOC)

4) Blood Vessels → PGE₂ and PGI₂ cause vasodilation
→ Iloprost [PGI₂] → Used for pulmonary HTN

5) Stomach. → PGE₂– Inhibit Proton pump
→ ↑ Mucous and bicarbonate

For NSAID Induced Peptic Ulcer Disease

Most specific drug is Misoprostol → DOC is proton pump inhibitors

6) Uterus. → PGE₂ → Relaxes Lower segment of uterus
→ Misoprostol [PGE₁ analogue]

7. Eye → PGF₂α → ↑ Uveo scleral outflow

Non-Selective COX Inhibitors Drugs

- Aspirin : Irreversible & ↑ uric acid
- Paracetamol [Acetaminophen] : COX 3 inhibitor, safe in CKD
- Ibuprofen : DOC for PDA
- Diclofenac : Partial COX-2 selective
- Indomethacin : Sedation is adverse effect
- Piroxicam : Longest acting NSAID

Selective COX inhibitors → celecoxib, parecoxib

Paracetamol Toxicity

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Analgesic of choice in renal disease

acetaminophen metabolite (NAPQI) depletes glutathione and forms toxic tissue byproducts in liver

– Antidote of paracetamol overdose- N-Acetyl Cysteine (DOC)

Aspirin poisoning (Salicylism)

NEETPG 23

Respiratory center stimulation

↓ Hyperventilation

↓

↓ CO₂

↓

Respiratory Alkalosis (Reversible)

↓

Metabolic Acidosis (Irreversible)

DOC → sodium bicarbonate (NaHCO_3)

GOUT

Acute Gout

1. NSAIDS [DOC]

2. Steroids
3. Colchicine

Mechanism of Colchicine

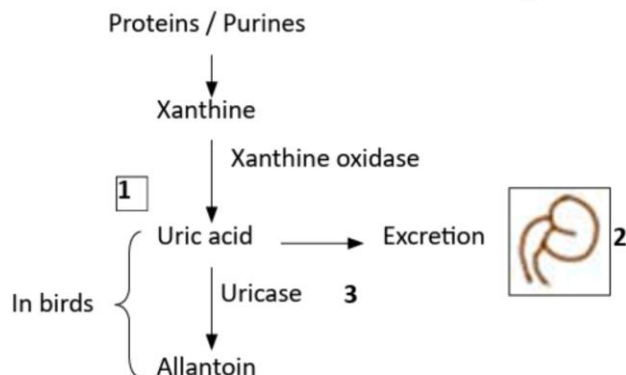
- Inhibit granulocyte migration
- Inhibit the release of glycoprotein from neutrophils

Chronic Gout

Three types of drugs can be used in chronic gout.

- Drugs decreasing production of uric acid
- Drugs increasing excretion of uric acid (Uricosuric agents)
- Drugs increasing metabolism of uric acid

NEETPG 25,23,FMGE 23,22



Mechanism of action	Drugs and its action
1.↓ Formation of Uric Acid.	Allopurinol → DOC for chronic gout Inhibit xanthine oxidase Febuxostat - Inhibit xanthine oxidase
2.↑ Excretion of Uric Acid [Uricosuric Drugs]	Probenecid Lesinurad → Plenty of fluids should be taken
3.↑ Uric Acid Metabolism (Rasburicase)	→ Recombinant uricase Pegloticase → oxidises uric acid → Long acting uricase These are fastest urate lowering agents for tumor lysis syndrome

RHEUMATOID ARTHRITIS

1. Conventional DMARD:

C	Chloroquine
P	Penicillamine
A	Azathioprine
G	Gold salts

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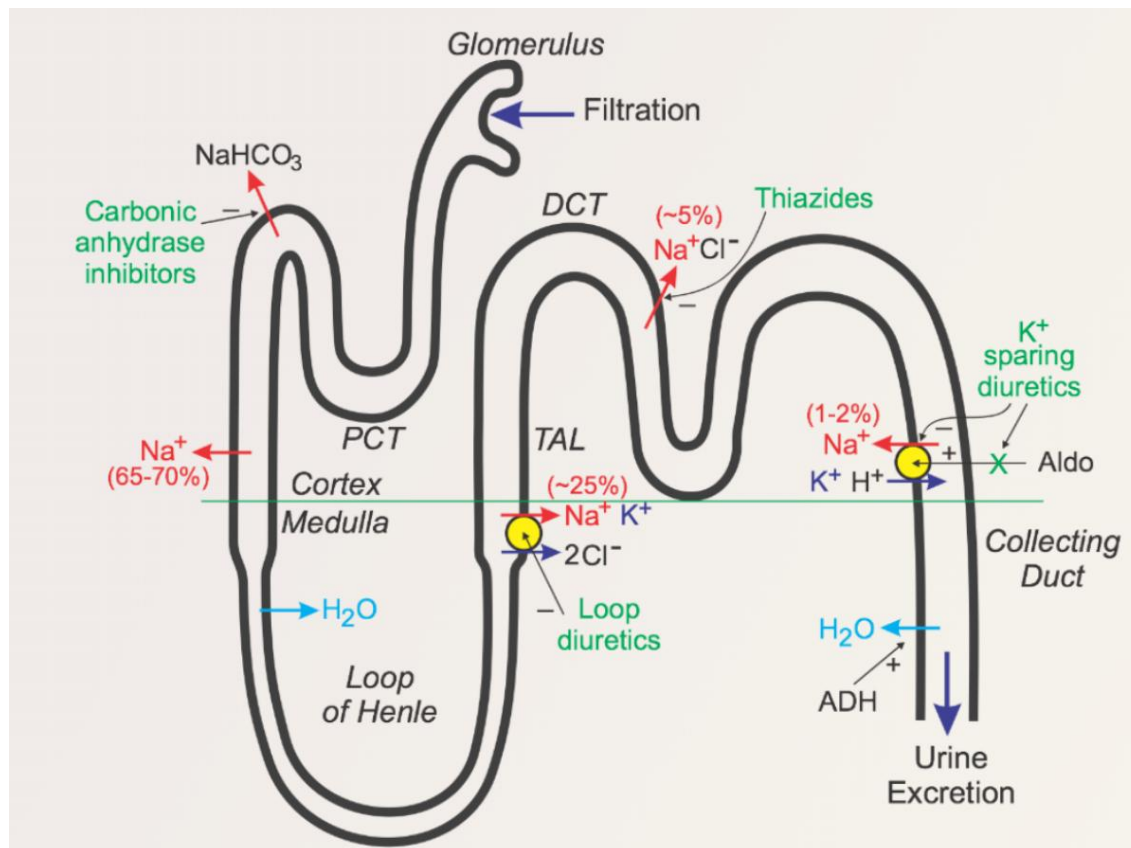
L	Leflunomide
I	Inhibitors of JAK
Malika	Methotrexate
Sherawat	Sulfasalazine

DMARD of choice in pregnancy → **Methotrexate**

2. Biological DMARD:

- Drugs Inhibiting TNF- α : Adalimumab -Subcutaneous route [S.C]
- IL - 1 Receptor Antagonists: Anakinra
- IL - 6 Receptor Antagonists: Tocilizumab → 1st IL- 6 targeted monoclonal Antibody
Sarilumab → Used for rheumatoid arthritis

DIURETICS



FMGE 25,24,23

Group	Site	Drug	Mechanism	Uses
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CA inhibitors	Proximal tubule	Acetazolamide Dorzolamide Brinzolamide		Angle closure glaucoma Doc- Mountain sickness
Loop diuretics	Thick ascending loop	Furosemide Torsemide Bumetanide	Na-k-2Cl	Hypertensive emergency – Bromide and iodide poisoning – Hypercalcemia
Thiazide	DCT	Methiazide Polythiazide Chlorthiazide	Na-Cl	HTN Renal stones
eNa channel blocker	Collecting duct	Amiloride	Na channel blocker	Diuresis Hyperkalemia Metabolic Acidosis
Aldosterone receptors antagonist	Collecting duct	Spironolactone Eplerenone	Aldosterone receptors blocker	DOC in - Conn's Syndrome Edema in Cirrhosis

Frequently asked points✓

1.Osmotic diuretics → Contra-Indications:

- Cerebral hemorrhage
- Acute renal failure
- Pulmonary edema

2. CARBONIC ANHYDRASE INHIBITORS → Adverse Effects:

- Metabolic Acidosis
- Hypokalemia
- Renal Stones

3.S/E of both thiazides and loop diuretics

↓ Na⁺ ↓ Mg²⁺ ↓ K⁺ ↓ H⁺
 ↑ Uric Acid ↑ Lipids. ↑ Glucose

NEETPG 24

CONGESTIVE HEART FAILURE

ACUTE CHF- Aim is to ↓ fluid and ↑ contractility of heart by giving diuretics and inotropes

CHRONIC CHF- Aim is to ↓work, ↓LVH, ↓ fluid, inotropic action
Drugs in chronic CHF- Diuretics ,Dilators, Decrease LVH, Digoxin

DILATORS

Venodilators - ↓ Preload - Nitrates

Arteriodilators - ↓ Afterload - Hydralazine

Both venous and arterial dilators -↓ both preload and afterload - Na nitroprusside, ACE inhibitors.

Diuretics- Fluid ↓

↓ **LVH remodeling-** by blocking Aldosterone

NEETPG 24,FMGE 22

These drugs decrease mortality in CHF

1. Beta Blockers
2. ACE inhibitors
3. Angiotensin Receptor Blockers
4. Aldosterone Antagonists
- 5.SGLT-2 inhibitors

DRUGS CAUSING GYNecomastia

DI : Digoxin S : Spironolactone C : Cimetidine, Ketoconazole O : Oestrogens

Digitalis / Cardiac Glycosides

–Can be given orally

–Used only in chronic CHF (**not acute**)

MOA in CHF Digitalis inhibits Na⁺K⁺ ATPase

↓ NCX Inhibition

↓

↑ Ca²⁺ in cytoplasm

↓

↑ Ca²⁺ in ER

↓

↑ Contractility

Frequently asked points✓

Digoxin → Only inotropic drug that can be given Orally

→ Adverse Effects

1.Nausea, Vomiting [MC]

2.Arrhythmias (MC arrhythmia → Ventricular bigeminy)

Most specific / characteristic → NPAT with AV Block

3.Xanthopsia / Yellow Vision

4.Gynaecomastia

DIGOXIN TOXICITY

NEETPG 22

Metabolic ↑ **Ca²⁺** ↓ **K⁺** ↓ **Mg²⁺**

Drugs → Quinidine, Verapamil, Amiodarone, Thiazides and Clarithromycin

DOC for Digitalis induced arrhythmias → Lignocaine/Phenytoin

Digibind for Severe poisoning

OTHER DRUGS FOR CHF

1.BNP (Brain Natriuretic Peptide) → Cause Natriuresis (↓Na)

→ Cause vasodilation

2.Vasopressin Receptors Antagonists: Conivaptan & Tolvaptan

Frequently asked points✓

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Funny Current Blocker

Ivabradine – Acts by blocking funny current (If) in S.A node by blocking Na channels

– S/E: ↓ in Visual Acuity

Hypertension

First line drugs

A- ACE inhibitors or ARBs

C- Calcium channel blocker

D- Diuretics

ACE INHIBITORS

FMGE 25,24

CAPTOPRIL LISINOPRIL ENALAPRIL RAMIPRIL

Frequently asked points✓

Inhibit bradykinin metabolism [↑Bradykinin]

→S/E →Dry cough and Angioedema

→C/I in pregnancy

→C/I in B/L Renal Artery Stenosis

→C/I in increased K⁺

→ Lower the risk of Diabetic Nephropathy

ARBs [Angiotensin (AT₁) Receptors Blockers]

NEETPG 23, FMGE 25

LOSARTAN → act as thromboxane A₂ and inhibit platelet aggregation

VALSARTAN

TELMISARTAN → Also stimulates PPAR → Used to reverse insulin resistance

ARBs do not cause cough and angioedema

Resistant hypertension

FMGE 25

When BP is not controlled by all 3 first line drugs including Diuretics

Doc: Spironolactone/ Epleronone

Doc for Hypertensive emergency → NICARDIPINE

Doc for Resistant HTN → SPIRONOLACTONE

HTN in Pregnancy → iv LABETALOL (block both α and β receptors) FMGE 24,23,26

ANTI-ARRHYTHMIC DRUGS

Class I → Na⁺ Channel Blockers

Class II → beta Blockers

Class III → K⁺ channel blockers

Class IV → Ca²⁺ Channel Blockers

Class V → Others

Class 1a Drugs	Class 1b Drugs	Class 1c Drugs
→ Quinidine → Procainamide	Lignocaine Phenytoin Tocainide	→ Encainide → Flecainide
Cause QT Prolongation	→ Used only for Ventricular arrhythmia → Lignocaine is the DOC for digoxin induced ventricular arrhythmias	Used for WPW syndrome Treatment of choice for WPW syndrome is radiofrequency ablation of aberrant pathway

Amiodarone

NEETPG 25,23, FMGE 24

→ Longest acting [t_{1/2} → > 3 weeks] anti-arrhythmic drug

→ MOA

1. Na⁺ channel Blocker
2. β -Blocker (Non-competitive)
3. K⁺ channel Blocker [Main action]
4. Calcium channel blocker

Act on phase 3 of cardiac action potential

Frequently asked points✓

Adverse effect of amiodarone:

Thyroid (hypo/hyper)
Peripheral neuropathy
Myocardial depression
Lung fibrosis & Liver toxicity
Corneal deposits
Photosensitivity (Rash on exposure to sun)

Digoxin - Used for AF (in patients with CHF)

Atropine - DOC for Bradycardia & AV block

Adenosine

NEETPG 25

Shortest acting antiarrhythmic drug ($t_{1/2} < 10s$)
DOC for PSVT
Prophylaxis for PSVT - oral Verapamil

Magnesium sulfate

DOC for Long QT Syndrome/Torsades De Pointes

Anti-Dyslipidemic Drugs

1. Statins

FMGE 24

Atorvastatin

Rosuvastatin [Longest Acting]

Pravastatin

MOA → Inhibit HMG-COA Reductase

Statins have maximum LDL – Cholesterol lowering potential

Adverse effects

Myopathy → Risk further ↑ with Fibrates and enzyme inhibitors like clarithromycin and erythromycin

Hepatotoxicity

Hyperglycemia: ↑ risk of new onset DM

2. Fibrates

FMGE 24

Clofibrate Fenofibrate Gemfibrozil

→ Fibrates have maximum TG lowering potential

Side effects:–

→ Fibrates increase risk of gall stones

→ Gemfibrozil can cause myopathy

3. Bile Acid Binding Agents (BABA)

Cholestyramine

Cholestipol

Cholesevalam

4. Intestinal Cholesterol Absorption Inhibitor [Ezetimibe]

– Inhibit NPL1C1 in the intestine (so cholesterol can't be absorbed)

5. Niacin (Vitamin B3)

NEET PG 24

– Has Maximum HDL-Cholesterol Increasing Potential

Frequently asked points✓

Adverse effects are flushing and itching (due to release of PGD2)

Hyperuricemia

Hepatotoxicity

Insulin resistance

6. PCSK9 inhibitor → Evolocumab

NEET PG 22

ENDOCRINE SYSTEM

1. Growth Hormone Inhibiting Hormones [GHIH] / Somatostatin

S → Secretory Diarrhoea

O → Oesophageal Varices

M → Malignancy [Islet cell tumors]

A → Acromegaly

2. Prolactin Inhibiting Hormone [PIH] = Dopamine [DA]

NEET PG 22

Bromocriptine

Cabergoline [Long acting]

Frequently asked points✓

NEET PG 22 FMGE 26

Drugs for Acromegaly

Gh receptor antagonist

Decrease GH release

Pegvisomant.

octreotide & cabergoline

Suppression of Lactation

– Prolactin is a milk secreting hormone

– Decreasing the level of prolactin can suppress lactation

3. Gonadotropin Releasing Hormone [GnRH]

GnRH Agonists

1. Leuprolide

GnRH Antagonists

1. Ganirelix

- 2. Nafarelin (nasal route)
- 3. Goserelin & Busurelin

- 2. Abarelix
- 3. Degarelix

Posterior Pituitary

- Secretes two main hormones
- Oxytocin – Vasopressin

Oxytocin → Main function is to contract the uterus

- DOC for augmentation of labour
- DOC for postpartum hemorrhage
- Ejection of milk
- DOC for Breast engorgement

Vasopressin/Antidiuretic hormone

Vasopressin Receptor Agonists

- V1 agonist is terlipressin
- DOC for esophageal varices

- V2 agonist is desmopressin
- DOC for neurogenic (central diabetes insipidus)

Frequently asked points✓

DOC for neurogenic DI is Desmopressin

DOC for nephrogenic DI is Thiazides

Vasopressin Receptor Antagonists

Vaptans → Conivaptan & Tolvaptan

Uses of vaptans

- 1) Congestive heart failure
- 2) SIADH

SIADH (Syndrome of Inappropriate ADH Secretion)

DOC are Vaptans (Conivaptan/Tolvaptan)

FMGE 25

THYROID

HYPERTHYROIDISM → Characteristic symptom (Heat intolerance, weight loss, sympathetic features)

Drugs used are:

Thyroid peroxidase inhibitors

Carbimazole & Methimazole (active)

- More potent
- More plasma t $\frac{1}{2}$
- Cross placenta easily

Propyl Thiouracil (PTU)

- Less potent
- Less plasma t $\frac{1}{2}$
- DOC in 1st trimester of pregnancy

Secretion inhibitors

1. NaI
 2. KI
 3. Lugol's Iodine
- Fastest acting anti-thyroid drugs

Peripheral conversion inhibitors

1. Propranolol
2. PTU
3. Prednisolone

Adjuvant drugs

Most important β # → Propranolol (along with treating the symptoms, it also ↓ peripheral conversion of T4 → T3)

Emergency hyperthyroidism →

D.O.C for thyroid storm : Propranolol

Antithyroid DOC : Propylthiouracil

ENDOCRINE PANCREAS

GLUCAGON

1. Hypoglycemia

MOA → acts by glycogenesis

Acts by Glycogenolysis

2. β - Blocker Poisoning

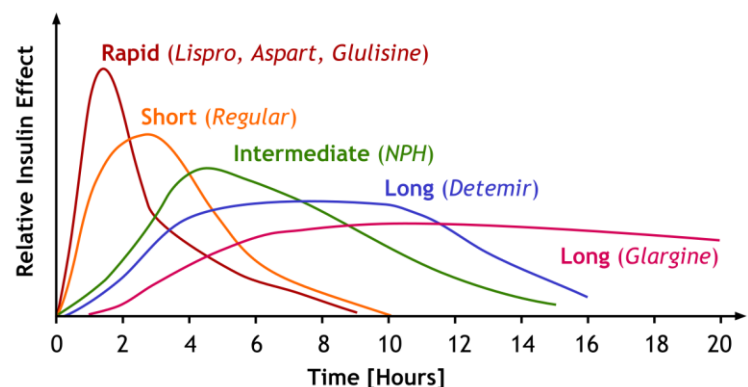
→ Glucagon is DOC

→ Acts by stimulating glucagon receptors on heart

INSULIN

Indications

1. All patients of type 1 DM
2. Uncontrolled patients of type 2 DM
3. Diabetes in pregnancy
4. Diabetic Ketoacidosis
5. Acute hyperkalemia



Side Effects of INSULIN

1. Hypoglycemia → MC and dangerous side effect which can be easily prevented
2. Hypokalemia

3. Lipodystrophy (Results from repeated injections at the same site)

ORAL ANTI DIABETICS DRUGS

FMGE 24,26

1. Insulin Secretagogues (Act by ↑ Insulin secretion)

1st Generation Chlorpropamide Tolbutamide

2nd Generation Glipizide Glimepiride Glibenclamide

Meglitinides Nateglinide Repaglinide

Side effect of Sulphonylureas

→ Cause hypoglycemia → Cause weight gain

2. AMP Kinase stimulators

→ Group k/a Biguanides

– Metformin – Phenformin

→ These drugs do not release insulin → so do not cause hypoglycemia

→ S/E of biguanides

1. Megaloblastic Anemia → More with Metformin

2. Lactic acidosis → More with Phenformin

3. Unexplained diarrhea → with metformin

Biguanides act by activating an enzyme AMP Kinase that results in

→ Gluconeogenesis (-)

→ Glycogenolysis (-)

→ Glycogenesis (+)

→ Glycolysis (+)

FMGE 23

3. Glitazones (Thiazolidinediones)

E.g. Troglitazone Rosiglitazone Pioglitazone

→ Mechanism Act by stimulating PPAR - γ

→ Reversal of Insulin Resistance

Adverse Effects

1. Hepatotoxic → Max. hepatotoxicity

2. Na⁺ and water Retention → Avoid in CHF and HTN

3. Predispose to osteoporotic fractures.

4. ↑ Risk of MI by ROSIGLITAZONE

5. ↑ Risk of urinary bladder carcinoma by Pioglitazone

4. Incretin-mimetic drugs**A. GLP analogues**

GLP-1 analog → Exenatide , Liraglutide

GLP-2 analog → Teduglutide

Side effects

- Acute pancreatitis (Major side effect)
- Nausea (Most common)

B. DPP - 4 inhibitors:

Sitagliptin Vildagliptin Alogliptin

Gliptins are oral anti diabetic drugs.

These do not cause weight gain or hypoglycemia

Side effects:

- **Nasopharyngitis (Most common)**
- **Pancreatitis**

5. SGLT-2 Inhibitors: → SGLT-2 inhibitors stop the reabsorption of glucose in PT resulting in glucosuria.

Drugs are

- Canagliflozin – Dapagliflozin
- Empagliflozin – Ertugliflozin

Frequently asked points✓

The most common side effect of SGLT-2 Inhibitors is Urinary Tract infections (urosepsis) and genital Tract infections.

SGLT-2 inhibitors provide cardiovascular benefits. These are approved for treatment of CHF.

Frequently asked question

Q) A 60-year-old female with type 2 diabetes mellitus was taking antidiabetic drugs and telmisartan. She has HbA1c level of 8.5. Her blood pressure is normal and recent investigations show eGFR to be 50 mL/min/1.73 m², but serum potassium is normal and she is asymptomatic. What should be done to decrease the risk of further decrement in eGFR?

- A. Increase the dose of telmisartan
- B. Shift to ramipril
- C. Increase the dose of anti-diabetic drug

D. Start carvedilol

Q) A patient of diabetes insipidus was given hydrochlorothiazide. It decreased urine output instead of acting as a diuretic. This paradoxical effect is due to which of the following mechanisms?

A. Direct stimulation of V2 receptors in the collecting duct

B) Inhibition of Na reabsorption in the distal convoluted tubule leading to mild volume depletion and increased proximal water reabsorption

C. Blocking aquaporin-2 channels in the collecting duct

D. Down regulation of V2 receptors of vasopressin

Q) 58 year old male with chronic kidney disease present for routine follow up. His bp is 140/100mmHg. Urine analysis reveals significant proteinuria. Which of the following is the most appropriate first line antihypertensive agent for this patient?

A) Propanolol

B) Hydrochlorothiazide

C) Telmisartan

D) Spironolactone

Adrenal

FMGE 23

Maximum glucocorticoid activity- Dexamethasone

Maximum mineralocorticoid activity -Aldosterone

Glucocorticoid with max mineralocorticoid activity - Hydrocortisone

Mineralocorticoid with max glucocorticoid activity - Fludrocortisone

Antenatal steroid use:

Betamethasone → IM 12 mg every 24 hrs. x 2 Doses

Dexamethasone → IM 6 mg every 12 hrs. x 4 Doses

OSTEOPOROSIS

Drug used in osteoporosis

Inhibit osteoblast	Calcitonin, Estrogens, SERM, Denosumab - Monoclonal antibody against RANK-Ligand
Stimulate osteoblast	PTH1-34
Dual action	Strontium

Bisphosphonates

Alendronate, Risedronate

Given orally Zoledronate → Given IV once yearly

- Inhibit osteoclasts
- DOC for Osteoporosis [Osteoporosis due to any reason]
- Highly toxic to esophagus

Frequently asked points✓

Side Effects of Bisphosphonates

- Esophageal toxicity/ esophageal ulcers
- Osteonecrosis of mandible

CNS

1.ALZHIEMER DISEASE

Mechanism	Drugs
AchE	Donepezil Rivastigimine (patch) Galantamine
NMDA antagonist	MEMANTINE
MAB	ADUCANUMAB

2.PARKINSON DISEASE

FMGE 25,22,26 NEET PG

22

- Dopamine precursor - L-dopa
- MAO-B inhibitor - Selegiline
- COMPT inhibitor - Entacapone
- Drug releasing DA - Amantadine → s/e - livedo reticularis
- Dopamine agonist - Pramipexole , Ropirinole → DOC
- Anticholinergic - Benzhexol,Benztropine → DOC in drug induced parkinsonism

Frequently asked question:

Q)Which of the following best describes the mechanism of action of memantine when used for Alzheimer's disease?

- Acetylcholinesterase inhibition
- NMDA receptor antagonist
- Blockade of A-beta amyloid
- Dopamine D2 receptor antagonism

ANTIEPILEPTIC DRUGS

FMGE 23,26NEETPG 22

Seizure	Drug
GTCS	Lamotrigine, valproate
Absence	Valproate, Ethosuximide
Focal seizure	Carbamazepine (DOC) Oxcarbazepine Levetiracetam (DOC in elderly)
Febrile seizure	Diazepam
Status Epilepticus	Lorazepam
Infantile spasms	ACTH (DOC)

VIGABATRIN

GABA transaminase inhibitor

Uses → Infantile spasm

S/E → visual field defects

PHENYTOIN

FMGE

25

“GUN HOT Z”

GUN → Gingival hyperplasia, Used long-term → Neuro (nystagmus/ataxia)

HOT → Hirsutism, Osteomalacia, Teratogen

Z → Zero-order kinetics

Sodium Valproate

MOA

Na⁺ Channel Blocker

Ca²⁺ channel Blocker

↑ GABA ↓ Glutamate

DOC for

- GTCS
- Absence Seizures
- Myoclonic Seizures
- Atonic seizures
- Lennox Gastaut syndrome
- Also used for Bipolar Disorder and prophylaxis of migraine

Side effect

- Pancreatitis, ↑ risk of PCOD (gender Specific S/E)
- Teratogenicity (most teratogenic Antiepileptic drug)
- Liver disease (hepatotoxicity: in young children has high incidence)

LAMOTRIGINE

- DOC for focal seizures in elderly (along with levetiracetam)
- Acts by Blocking Na⁺ channels, Increasing GABA and Decreasing glutamate activity
- Side effects:
 - Steven Johnson Syndrome
 - Toxic Epidermal Necrolysis

Frequently asked question

Q) A 10-year-old boy with generalized tonic-clonic seizures has been on multiple antiepileptic drugs for the last 3 weeks. He now presents with high-grade fever, painful erythematous rash, and petechial lesions on the lips as seen in figure below. There is concern for a serious drug-induced cutaneous reaction. Which of the following antiepileptic drugs can be safely continued in this child?

- A. Phenytoin
- B. Valproate
- C. Lamotrigine
- D. Clobazam

CLOZAPINE → DOC for resistant Schizophrenia

Side effects are:

- Agranulocytosis [Dose independent]
- Seizures [Dose dependent]
- Sedation (Most common adverse effect)
- Sialorrhea (MC side effect)

Epilepsy in Pregnancy:

- Most teratogenic: Valproate
- Least teratogenic: Lamotrigine, Levetiracetam

New Antiepileptic Drugs

1. Bind to SV2A:

Levetiracetam

Brivaracetam

- These are DOC for focal seizures in elderly

- S/E – CNS side effects like aggression, irritability and mood changes.

2. Cannabidiol:– Stimulate CB1

– Approved for Dravet syndrome

3. Stiripentol:– Increase the action of GABA

– Inhibits the enzyme LDH

LITHIUM

FMGE 24,23,26

Half-life is 24 hours

Narrow Therapeutic Index

Therapeutic drug monitoring → Plasma concentration of Lithium

Acute mania → 0.8-1.2 mEq/L

Prophylaxis → 0.5-0.8 mEq/L

Side effects - Mnemonic: “LITHIUM”

Leukocytosis

Increased urination (polyuria)

Tremor / Thyroid (↓ function)

Heart defect (Ebstein anomaly)

Insipidus (nephrogenic)

Upset stomach

Mass gain (weight gain)

Frequently asked question:

Q)Which of the following ECG change is most characteristically associated with lithium therapy?

A. ST-segment elevation

B. ST-segment depression

C. QT interval prolongation

D.T-wave flattening

DRUG OF ABUSE

1.OPIOIDS

NEETPG 24,22,FMGE 25

MORPHINE

Acts on μ , K , δ Receptors

μ Receptor Functions

S - Sedation A – Analgesia C – Constipation

R - Respiratory depression U - eUphoria M – Miosis

Kappa receptors cause dysphoria

DOC for acute Opioid poisoning → Naloxone

DOC for maintenance in opioid poisoning → Naltrexone

Commonly used maintenance therapy in opioids withdrawal → buprenorphine

Drugs causing Disulfiram Like Reaction

C - Cefoperazone & Chlorpropamide

G - Griseofulvin

M - Metronidazole

P - Procarbazine

Drugs ↓ alcohol craving:

Naltrexone

Ondansetron

Opiramate

Acamprosate

ANTI PLATELET DRUGS

FMGE 24

A – Aspirin → Blocks COX-1/COX-2 → ↓ TXA₂ → ↓ platelet aggregation

C – Clopidogrel (also Prasugrel, Ticagrelor) → Blocks P2Y₁₂ (ADP receptor) → ↓ GpIIb/IIIa activation

I – Ilprost / Epoprostenol → Prostacyclin analogs → ↑ cAMP → ↓ platelet aggregation

A – Abciximab (also Eptifibatide, Tirofiban) → Blocks GpIIb/IIIa receptor → prevents fibrinogen binding

P – Phosphodiesterase inhibitors (Dipyridamole, Cilostazol) → ↑ cAMP → ↓ platelet aggregation

C – Cangrelor → IV P2Y₁₂ inhibitor (short-acting)

HEMATOPOIETIC GROWTH FACTORS

NEETPG 23, FMGE 23

Cells	Growth factor	Drugs
RBC	Erythropoietin	Darbopoietin
WBC	G-CSF GM-CSF	Filgrastim Sargramostim
Platelets	IL-11	Oprelvekin

ANTI COAGULANTS DRUGS

NEETPG 24, FMGE 24,23

Types	Route	Examples	MOA	Antidote
Unfractionated Heparin (UFH)	IV / SC	Heparin	Activates Antithrombin III → inhibits Thrombin (IIa) and Factor Xa	Protamine sulfate
Low Molecular Weight Heparin (LMWH)	SC	Enoxaparin	Same as heparin but mainly inhibits Factor Xa	Protamine (partial reversal)
Fondaparinux	SC	Fondaparinux	Selectively inhibits Factor Xa via Antithrombin III	-
Direct Thrombin Inhibitors (DTIs)	IV / Oral	Argatroban, Bivalirudin, Dabigatran (oral)	Directly inhibit Thrombin (IIa)	Idarucizumab (for Dabigatran)
Vitamin K Antagonist	Oral	Warfarin	Inhibits Vitamin K epoxide reductase → ↓ Factors II, VII, IX, X	Vitamin K, Fresh Frozen Plasma (FFP), PCC
Direct Factor Xa Inhibitors (DOACs)	Oral	Rivaroxaban, Apixaban, Edoxaban	Directly inhibit Factor Xa	Andexanet alfa

PEPTIC ULCER DISEASE

Most specific drug for NSAID induced Peptic ulcer - misoprostol

DOC for NSAID induced Peptic ulcer- PPI

PPI-

DOC for PUD due to any reason

DOC for GERD

DOC for Zollinger Ellison Syndrome

Antacid → Fastest pain relievers of PUD

H pylori

Triple drug therapy - given for 2 weeks

C- clarithromycin A-amoxicillin P- pantoprazole

FMGE 25,26

Frequently asked question

Q) A 26yr old male presents to the hospital with fever and arthritis. Ecocardiography confirms the presence of carditis in this patient. He gives a past history of peptic ulcer disease. Which of the following is the most appropriate anti-inflammatory agent in this patient?

- a. Diclofenac
- b. Tramadol
- c. Celecoxib
- d. Ibuprofen

Q) Which of the following is correct treatment of peptic ulcer associated with H pylori?

- A) Omeprazole only
- B) Clarythromycin plus Amoxicillin
- C) Omeprazole, Clarythromycin and Amoxicillin
- D) Metronidazole, Omeprazole and Clarythromycin

Antimicrobial Agents

CIDAL: These kill the microorganisms

STATIC: These inhibit the growth of microorganisms

Beta lactams

VA-VAncomycin

F-Fluoroquinolones

A-Aminoglycosides

Protein synthesis inhibitors-

Bind AT 30S ribosomes: Aminoglycosides Tetracyclines

Aminoglycosides-

it is a cidal drugs

Cause misreading of mRNA

s/e-nephrotoxic, ototoxic

Tetracycline-

Prevent attachment of aminoacyl- tRNA

Resistance- Due to development of efflux pumps

DOC for cholera

Doxycycline- DOC for rickettsia

SELL at 50S- streptogramins, Erythromycin group, Linezolid and Lincosamides

Spectrum of Penicillins - Mainly against gram positive bacteria

Ampicillin is DOC for listeria

Cephalosporins

FMGE 24

1st Generation "FA/PHA"	2nd Generation "Everything Else"	3rd Generation "ONE/TEN/IME"	4th Generation "PI"	5th Generation "ROL"
* CEF A ZOLIN CEF A DROXIL * CEF H ALEXIN CEP H ALOTHIN CEP H APIRIN CEP H RADINE	CEFOXITIN CEFOTETAN CEFMETAZOLE CEFPROZIL CE F ACLOR CE F UROX IME	* CEF T RIAX ONE CE T IBUT EN * CEFOTAX IME * CE T AZID IME CE F PODOX IME CE F IX IME CEFDINIR MOXALACTAM	* CEF E PI ME CE F PI ROME	* CE T A R OLINE CE T OBI P RO LE

Surgical prophylaxis DOC- cefazolin

NEETPG 23

5th generation are effective against MRSA

Cefaparazone can not be given to alcoholic patient → it shows disulfiram like reactions

Macrolides

DOC of CLAP

C-chancroid

L-legionella

A-atypical pneumonia

P-pertussis

Fluoroquinolones

FMGE 25, NEETPG 22

Inhibition of DNA gyrase and topoisomerase IV

C/I in pregnancy and children (<18 yrs) [Cause cartilage and tendon damage]

Antimicrobials safe in pregnancy- P-penicillin C-clindamycin M- macrolide

Frequently asked question

Q)A 24-year-old otherwise healthy woman presents with 2 days of dysuria, increased frequency of micturition, and suprapubic discomfort. She has no fever, flank pain, vomiting, or vaginal discharge. Her vitals are stable and there is no history of recent antibiotic use or urinary tract abnormalities. She is diagnosed with uncomplicated lower urinary tract infection (UTI) and empirical therapy is being considered. Which of the following drugs should NOT be preferred as first-line empirical treatment in this case?

A) Nitrofurantoin

- B) Cotrimoxazole
- C) Ciprofloxacin
- D) Fosfomycin

Avoid in preg

Tetracycline-damage bone and teeth

Aminoglycoside-hearing loss

Fluoroquinolone-cartilage and tendon damage

Malaria :

Doc for plasmodium falciparum- Chloroquine

Quinine is safe in first trimester Side effect is bulls eye maculopathy

Artesunate is CI in pregnancy

Combination used in india North east state- ME ArteMEther +luMEfantrine

All over india Artesunate + sulfadoxine + pyrimethamine

Artemisinin CI in first trimester For 1st tri- chloroquine

PF malaria in 1st tri- quinine ACT is given in 2nd and 3rd trimester of pregnancy

Malaria prophylaxis

Short term < 6wk Doxycycline daily Started 2 days before to 4 weeks after the journey

Long term > 6wk Mefloquine Weekly Started 2 weeks before to 4 week after journey

Frequently asked question

Q)A 32-year-old man returned from a malaria-endemic area with high-grade fever for 5 days. On examination he is drowsy, confused, and has altered sensorium. Peripheral smear shows Plasmodium falciparum. What is the most appropriate treatment?

A. Intravenous artesunate

B. Intravenous artesunate with sulfadoxine-pyrimethamine

C. Oral artemether-lumefantrine

D. Oral chloroquine

Q)A 25-year-old man returns from (Manipur with high-grade fever and chills for 3 days. Peripheral smear shows Plasmodium falciparum trophozoites. What is the most appropriate treatment?

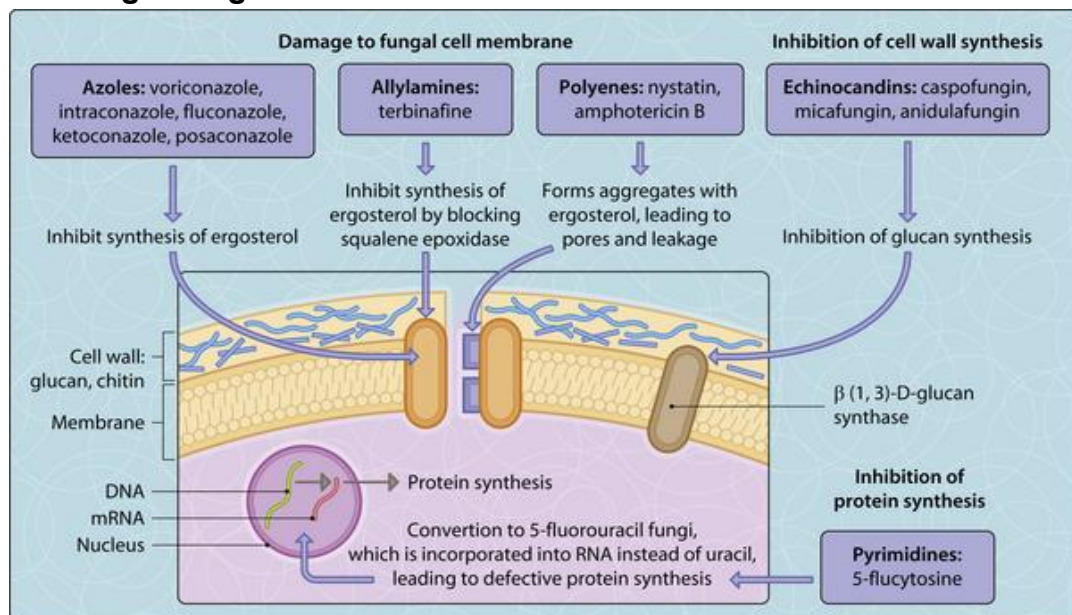
A. Artemether-lumefantrine (AL) for 3 days + single dose primaquine on Day 2

B. Artemether-lumefantrine (AL) for 3 days + primaquine for 14 days

C. Chloroquine for 3 days + primaquine single dose

D. Oral quinine for 7 days

Antifungal drugs



ANTICANCER DRUGS

1. Alkylating Agents

Cyclophosphamide	Cause Hemorrhagic cystitis and Pulmonary fibrosis
Busulfan	Cause pulmonary fibrosis
Mitomycin-C	Used topically for Subglottic stenosis

2. Antimetabolites (S-phase specific)

NEETPG 24, FMGE 24,26

a. Folic Acid Antagonist

Methotrexate → Inhibits dihydrofolate reductase

Uses:

Choriocarcinoma – DOC

Rheumatoid arthritis – DOC

Toxicity antidote: Folinic acid (Leucovorin)

Frequently asked question

Q) A 65-year-old woman with breast cancer was given combination chemotherapy including methotrexate. To reduce toxicity, leucovorin is administered. Which of the following enzymes is bypassed by leucovorin rescue during methotrexate therapy?

- A. Thymidylate synthase
- B. Dihydrofolate reductase
- C. Dihydropyrimidine dehydrogenase
- D. Methionine synthase

b. Pyrimidine Analogues

5-FU (5-Fluorouracil): Causes hand–foot syndrome

Cytarabine: Causes cerebellar toxicity

c. Purine Analogues

6-Mercaptopurine: Hepatotoxic

Cladribine: DOC – Hairy cell leukemia

Fludarabine: DOC – CLL

Bleomycin

FMGE 25,26

Marrow sparing

Causes pulmonary fibrosis

Causes flagellated (striped) skin pigmentation

Frequently asked question:

Q) Which of the following is adverse affect associated with bleomycin use?

- A) Bone marrow suppression
- B) Pulmonary fibrosis
- C) Cardiotoxic
- D) Renal failure

ANTIVIRAL DRUGS

ANTI HIV DRUGS

1. Fusion/Entry Inhibitors

Enfuvirtide → binds GP41 → blocks fusion (SC)

Fostemsavir → binds GP120 (oral)

Maraviroc → CCR5 antagonist (oral)

Ibalizumab → anti-CD4 monoclonal Ab (IV)

2. Reverse Transcriptase Inhibitors (RTI)

➤ NRTI (Competitive)

Drugs: Zidovudine, Lamivudine, Tenofovir, Abacavir, Emtricitabine

Key A/E:

Zidovudine → BM suppression

Abacavir → MI risk

Didanosine → pancreatitis

Stavudine → neuropathy

All → pancreatitis + neuropathy

Safest → Lamivudine

HBV drugs → L + E + T

► NNRTI (Non-competitive)

1st gen: Efavirenz, Nevirapine, Delavirdine

2nd gen: Rilpivirine, Etravirine, Doravirine

Points:

HIV-1 only

Resistance common → not used alone

CYP metabolism → drug interactions

Nevirapine → prevents vertical transmission, hepatotoxic

3. Integrase Inhibitors

Drugs: Dolutegravir, Elvitegravir

First-line drugs now

Elvitegravir + Cobicistat (CYP3A4 inhibitor booster)

4. Protease Inhibitors

Drugs: Ritonavir, Lopinavir, Atazanavir, Indinavir, etc.

Key points:

CYP3A4 metabolism

Ritonavir → strongest inhibitor (boosting drug)

A/E (Lipodystrophy):

↑ Glucose, ↑ Lipids

Insulin resistance

Weight gain

Treatment (HAART)

Start in all HIV patients

Lifelong therapy

Use ≥3 drugs (≥2 classes)

First-line:

2 NRTI + 1 NNRTI / Integrase inhibitor

Common:

T + L + E or T + L + D

HIV + TB

Start ATT first

Start ART after 2 weeks

Post exposure prophylaxis

→ **Should be started as early as possible after exposure (within maximum limit of 72 hours)**

→ Should be given for 28 days (4 weeks)

→ All the drugs are given orally → Drugs: Tenofovir + Lamivudine + Protease Inhibitors (or integrase inhibitors)

3. Prevention of vertical transmission:

→ Mother should be given full HAART therapy (TLE)

→ After delivery, Baby is given Nevirapine alone for 6 weeks [if mother is already exposed to Nevirapine, zidovudine is given]

ANTI-HEPATITIS VIRUS DRUGS

Hepatitis- B

→ D.O.C for chronic HBV is Tenofovir (1st priority)/Entecavir

→ Alternate to this, (Oral drugs effective against H.I.V and Hepatitis B)

- L - Lamivudine
- E - Emtricitabine
- T - Tenofovir

Hepatitis C

→ Previously treated with interferon and ribavirin

New Oral Drugs for Hepatitis C

Protease Inhibitors → Telaprevir ,Boceprevir

NS 5A Inhibitors → Elbasvir Ledipasvir

Antifungal Drugs – Very Short Summary

Mechanism (MOA)

- **Cell membrane (Ergosterol):**
 - **Polyenes** (Amphotericin B, Nystatin) → pores (fungicidal)
 - **Azoles** → ↓ ergosterol synthesis (fungistatic)
 - **Allylamines** (Terbinafine) → ↓ squalene epoxidase
- **Cell wall** → Echinocandins (↓ β -1,3-glucan)
- **DNA** → 5-Flucytosine
- **Mitotic spindle** → Griseofulvin

Polyenes

- **Amphotericin B**
- DOC: severe systemic fungal infections
- A/E: **nephrotoxic** (MC)
 - infusion fever/chills, BM suppression
- **Liposomal AMB** → less nephrotoxic
- **Nystatin** → topical Candida

Azoles (Mechanism, left)

- **Ketoconazole, Fluconazole, Itraconazole, Voriconazole**
- **MOA:** inhibit lanosterol-14 α -demethylase

Echinocandins

- **Caspofungin**
- Uses: Candida, Aspergillus
- **MOA:** inhibit cell wall

Griseofulvin

- **MOA:** Mitotic spindle inhibition
- Uses: **dermatophytes (tinea)**
- Alcohol → disulfiram-like reaction

High-yield extras

- Drugs causing **gynecomastia** → Ketoconazole
- Amphotericin B → “Ampho-terrible” (nephrotoxic)
- Dermatophytes → drugs concentrate in **keratin** (skin, hair, nails)

Azoles (right)

- Drugs: **Ketoconazole, Fluconazole, Itraconazole, Voriconazole, Posaconazole**
- Inhibit lanosterol-14 α -demethylase.
- **Key points:** ▼ **Ketoconazole**, → gynecomastia, adrenal suppression, hepatotox
- **Fluconazole** → best CNS penetration (Cryptococcus maintenance)
- **Itraconazole** → Histoplasma, Sporothrix, Blastomyces
- **Voriconazole** → DOC invasive aspergillosis
- **Posaconazole** → Mucormycosis (alternative)

5-Flucytosine

- **MOA:** inhibit DNA synthesis
- Use: with Amphotericin B (Cryptococcal meningitis)

Topical Antifungals

- **Azoles:** Clotrimazole, Miconazole
- **Others:** Ciclopirox, Amorolfine, Tavaborole
- **Nail lacquer:** Ciclopirox, Amorolfine, Efinaconazole, Tavaborole

Drugs and antidote**FMGE 23, NEET PG 22**

Acetaminophen	N-acetylcysteine
Aspirin	Sodium bicarbonate
organophosphates	atropine
Benzodiazepines	Flumazenil
Beta blockers	glucagon
Copper	“Penny”cillamine (penicillamine),
Cyanide	Hydroxocobalamin, nitrites + sodium thiosulfate
Dabigatran	Idarucizumab
Heparin	Protamine sulfate
warfarin	Vit k agonist
Opioids	naloxone
methotrexate	Folinic acid
alcohol	fomepizole

Condition and DOC**NEETPG 25, FMGE 24**

HTN in pregnancy	Oral labetalol
HTN emergency in pregnancy	IV labetalol
HTN	diuretics
HTN emergency	nicardipine
Bradycardia	atropine
PSVT	adenosine
Focal seizures	carbamazepine
Trigeminal neuralgia	carbamazepine

Focal seizures	Levetiracetam (in elderly)
Infantile spasms	ACTH
eclampsia	Mgso4
dystonia	benzhexol
akathisia	propranolol
Tardive dyskinesia	valbenazine
Malignant neuroleptic syndrome	Dantrolene
Opioid poisoning	naloxone
Maintenance in opioid poisoning	naltrexone
Heparin induced thrombocytopenia	bivalirudin
MRSA	vancomycin
VRSA	daptomycin
Pseudomembranous colitis	fidaxomicin
hepatitis	Tenofovir
Candidiasis	fluconazole
Aspergillosis	voriconazole
Mucormycosis, cryptococcal meningitis, kala azar	Amphotericin B
dermatophytosis	terbinafine
P vivax, P vivax in first trimester	chloroquine
P. falciparum	ACT
P. falciparum in first trimester	Quinine
Cerebral malaria	IV artesunate
Malaria in north northern states	Lumefantrine + artemether
Rest of india	Artesunate+sulfadoxine pyrimethamine
Radical cure of p. Vivax and malaria	Primaquine

Drug reactions	Causal agents
Diabetes insipidus	Lithium, demeclocycline
SIADH	Carbamazepine, Cyclophosphamide, SSRIs
Agranulocytosis	Dapsone, clozapine, carbamazepine, propylthiouracil, methimazole
Gray baby syndrome	Chloramphenicol
Hemolysis in G6PD deficiency	Sulfonamides, dapsone, primaquine, aspirin, nitrofurantoin
Gingival hyperplasia	Cyclosporine, Ca ²⁺ channel blockers, phenytoin
Stevens-Johnson syndrome	Lamotrigine
Teeth discoloration	Tetracyclines
Tendon/cartilage damage	Fluoroquinolones
Pulmonary fibrosis	Methotrexate, nitrofurantoin, bleomycin, busulfan, amiodarone, pirfenidone, methysergide

Miscellaneous

Benralizumab act on IL-5	NEETPG 25
Drug for HIV induced lipodystrophy - TESAMORELIN	NEETPG 24
Drug used in treatment of multiple myeloma associated with increased risk of developing herpes zoster infection → bortezomib	NEETPG 24
Aprepitant - NK-1 antagonist → use to prevent late onset chemotherapy-induced nausea and vomiting	NEETPG 24
Drug used for both HIV and hepatitis - tenofovir	FMGE 24
Drug for the treatment of post-menopausal breast cancer → anastrozole	
Dicumarol → inhibit vit k dependent clotting factors	FMGE 23